



# Harnessing Generative Models for Synthetic Non-Life Insurance Data

EAA e-Conference on  
Data Science & Data Ethics

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MOTIVATIONS

01

# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## MOTIVATIONS

- Insurance Use Case
- Employ a wide range of Generative Models



Google Nano Banana

USE CASE

02

## DATA SCARCITY IN INSURANCE RESEARCH

- **The Problem:** Access to realistic datasets is a major barrier to advancing actuarial research and developing insurance analytics tools.
- Most insurance data is confidential and proprietary
- Data masking and bureaucratic processes prevent disclosure
- Limited open resources lack diversity for robust modeling
- **The Solution:** Develop simulated datasets using generative models

## THE ANATOMY OF INSURANCE NON-LIFE RISK DATA

- **Risk Premium Components:**
  - **Frequency = Number of Claims / Number of Exposures**
  - **Severity = Losses / Number of Claims**
  - **Risk Premium = Frequency × Severity**
- *The risk premium equals the product of expected claim frequency and expected cost per claim*

## DATASETS EMPLOYED

- **Datasets are retrieved from the “CASdatasets” R Package**
- **Dataset 1:** ausprivauto0405 - Automobile claim datasets in Australia
- It is a data frame of 9 columns and 67,856 rows:
- Exposure, VehValue, VehAge, VehBody, Gender, DrivAge, ClaimOcc, ClaimNb, ClaimAmount
- **Dataset 2:** swmotorcycle - Swedish Motorcycle Insurance dataset
- It is a data frame of 9 columns and 64,548 rows:
- OwnerAge, Gender, Area, RiskClass, VehAge, BonusClass, Exposure, ClaimNb, ClaimAmount

## UNLOCKING DATA QUALITY: LEVERAGING CLAIM OCCURRENCE

- **What is ClaimOcc?**
  - It is a binary indicator that indicates the occurrence of a claim
  - If  $\text{ClaimOcc} = 0 \rightarrow \text{ClaimNb} = 0$  and  $\text{ClaimAmount} = 0$
  - If  $\text{ClaimOcc} = 1 \rightarrow \text{ClaimNb} \geq 1$  and  $\text{ClaimAmount} > 0$
- **Why use ClaimOcc?**
  - Structural relationship: drives logical dependency between ClaimNb and ClaimAmount
  - Class imbalance: separates rare claim events from common no-claim cases
  - Model stability: improves synthetic data quality by conditioning
- **Role in Generative Models**
  - **CGMM**: Fits separate Gaussian mixture models for each ClaimOcc value
  - **CVAE & CTVAE**: Concatenates ClaimOcc to the latent encoding for conditional generation
  - **CDM**: Uses ClaimOcc as a condition signal to guide the denoising process



## SYNTHETIC DATA GENERATION TRIALS

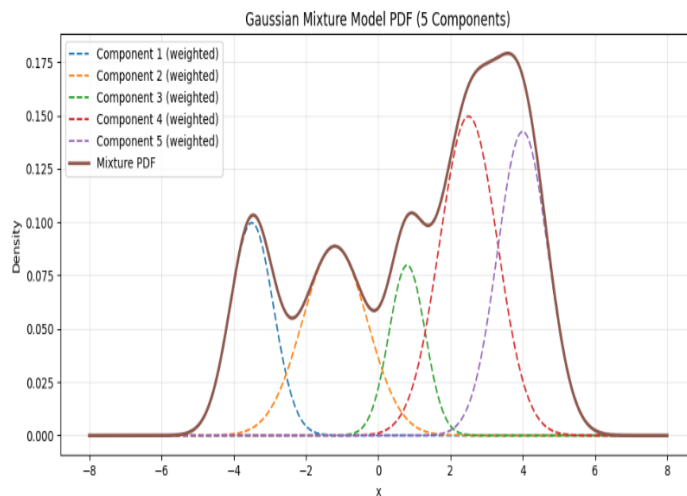
- **Trial 1: 80% Training → 80% Generation**
- **Baseline Quality Assessment**
  - What is the baseline quality of synthetic insurance data when a generative model is trained on 80% of the original data? How well does it preserve statistical distributions and enable accurate GLM predictions?
- **Trial 2: 60% Training → 80% Generation**
- **Data Augmentation with Limited Training**
  - Can a generative model trained on only 60% of the data successfully generate synthetic samples to reach 80% dataset size while maintaining statistical fidelity and predictive performance?
- **Trial 3: 80% Gender-Masked → 80% Generation**
- **Fairness-Aware Generation**
  - Can a gender-masked generative model generate unbiased synthetic insurance data while maintaining predictive utility, and what is the fairness-accuracy trade-off?

# CONDITIONAL GAUSSIAN MIXTURE MODELS

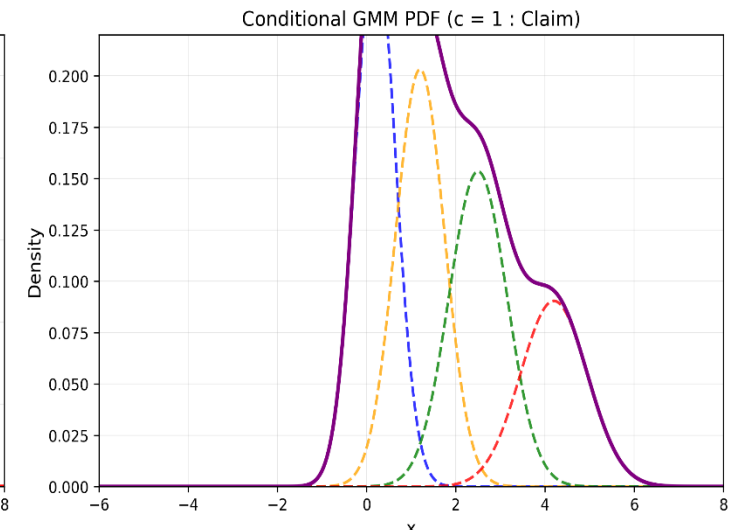
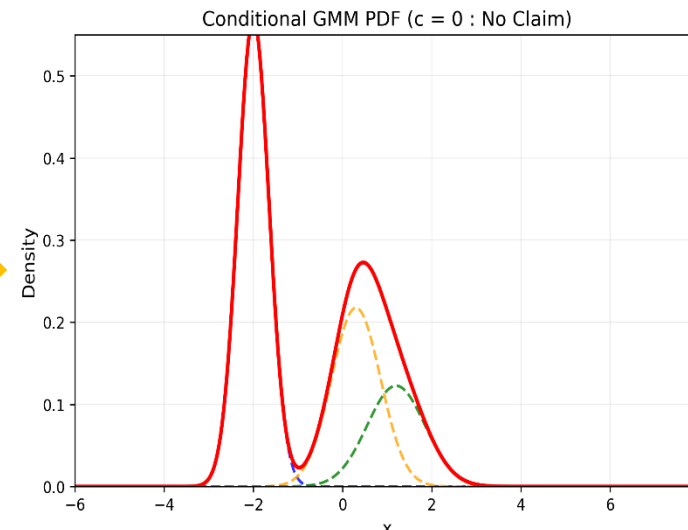
03

## CONDITIONAL GAUSSIAN MIXTURE MODELS (CGMM)

- **What is it?**
- A Gaussian Mixture Model assumes that the data are generated by a mixture of  $K$  Gaussian distributions, each with unknown parameters and corresponding to a cluster.
- Every Gaussian has its own mean  $\mu_k$ , covariance  $\Sigma_k$ , and mixing weight  $\pi_k$ .



ChatGPT



$$Loss = - \sum_{n=1}^N \log \left( \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right)$$



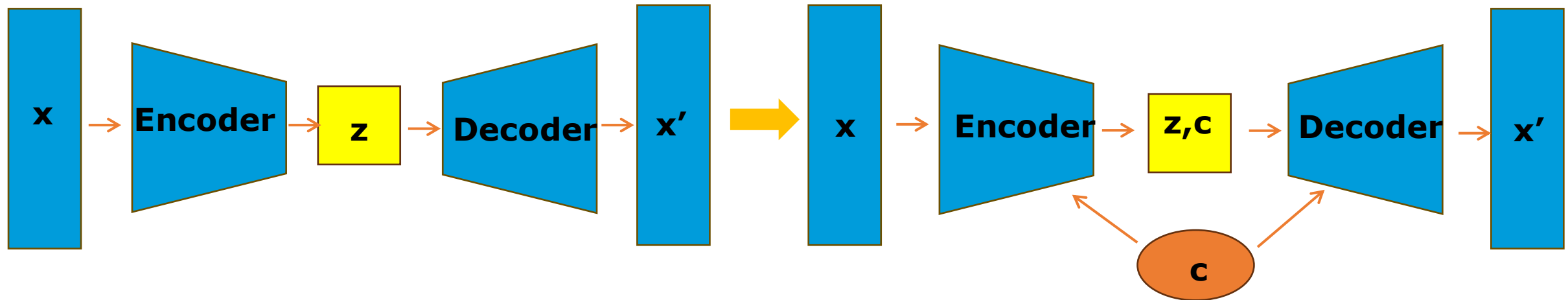
$$Loss = - \sum_{n=1}^N \log \left( \sum_{k=1}^K \pi_k(c_n) N(x_n | \mu_k(c_n), \Sigma_k(c_n)) \right)$$

# CONDITIONAL VARIATIONAL AUTOENCODER

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## CONDITIONAL VARIATIONAL AUTOENCODER (CVAE)

- **What is it?**
- A Variational Autoencoder (VAE) is a type of neural network that learns to represent input data as a probability distribution in a lower-dimensional space and then decode it to generate similar input data.



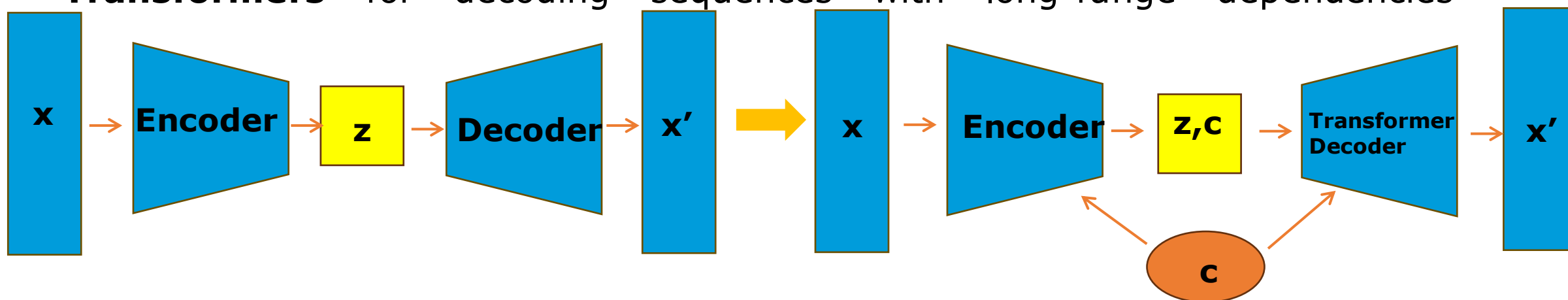
$$Loss = -E_{q_{\phi}(z|x)}[\log_{p_{\theta}}(x|z)] + KL(q_{\phi}(z|x)||p(z)) \longrightarrow Loss = -E_{q_{\phi}(z|x,c)}[\log_{p_{\theta}}(x|z,c)] + KL(q_{\phi}(z|x,c)||p(z,c))$$

CONDITIONAL  
VARIATIONAL  
AUTOENCODER  
WITH A  
TRANSFORMER  
BASED  
DECODER

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## CONDITIONAL VARIATIONAL AUTOENCODER WITH A TRANSFORMER-BASED DECODER (CTVAE)

- **What is it?**
- It's a hybrid architecture that leverages:
- **-VAEs** for learning a smooth, continuous latent space.
- **-Transformers** for decoding sequences with long-range dependencies



$$Loss = -E_{q_{\phi}(z|x)}[\log_{p_{\theta}}(x|z)] + KL(q_{\phi}(z|x)||p(z)) \xrightarrow{\text{yellow arrow}} Loss = -E_{q_{\phi}(z|x,c)}[\log_{p_{\theta}}(x|x_{<t}, z, c)] + KL(q_{\phi}(z|x, c)||p(z, c))$$

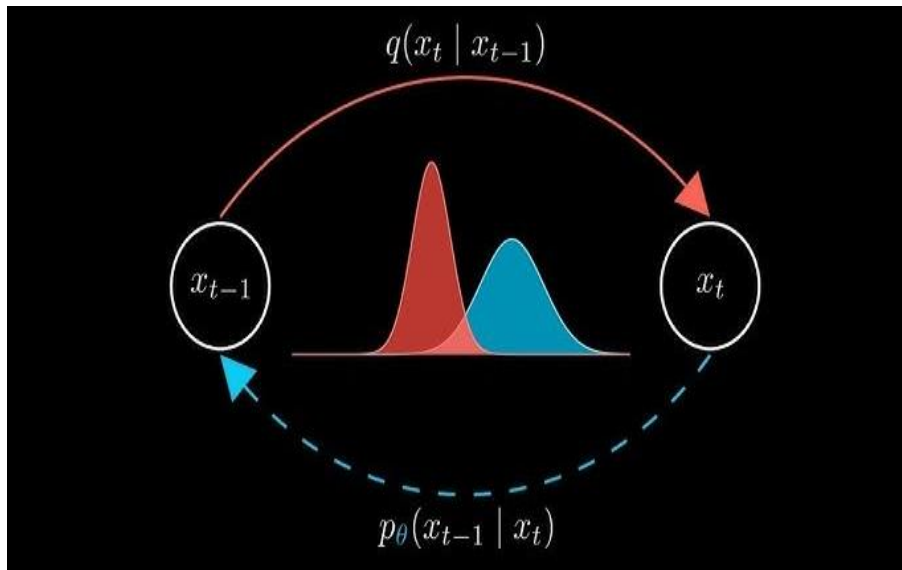
# CONDITIONAL DIFFUSION MODEL

06



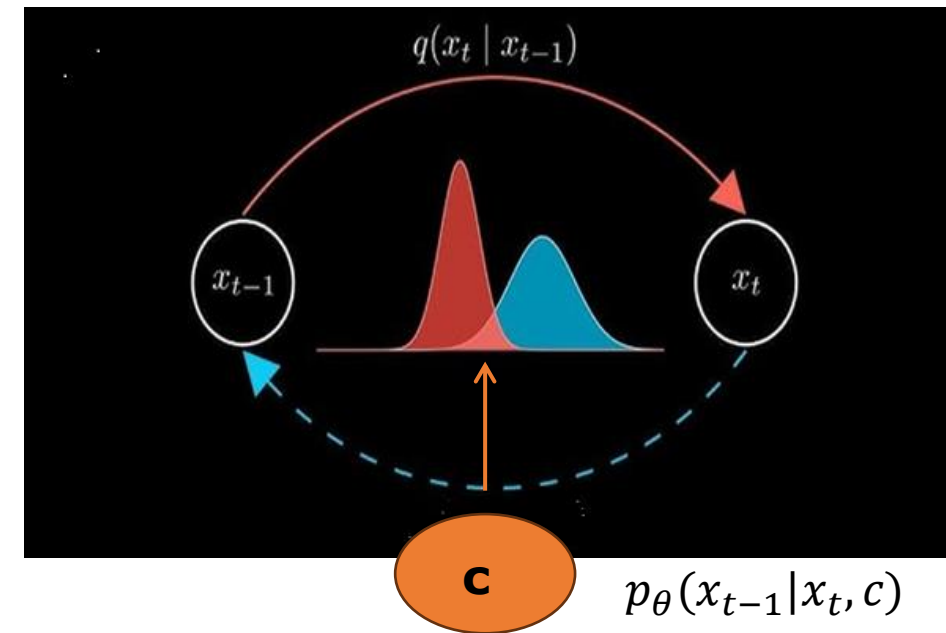
## CONDITIONAL DIFFUSION MODEL

- **What is it?**
- A Diffusion Model learns to generate data by simulating a gradual noising-and-denosing process.



$$Loss = E_{x_0, t, \epsilon} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

source



$$Loss = E_{x_0, t, \epsilon} [\|\epsilon - \epsilon_{\theta}(x_t, t, c)\|^2]$$

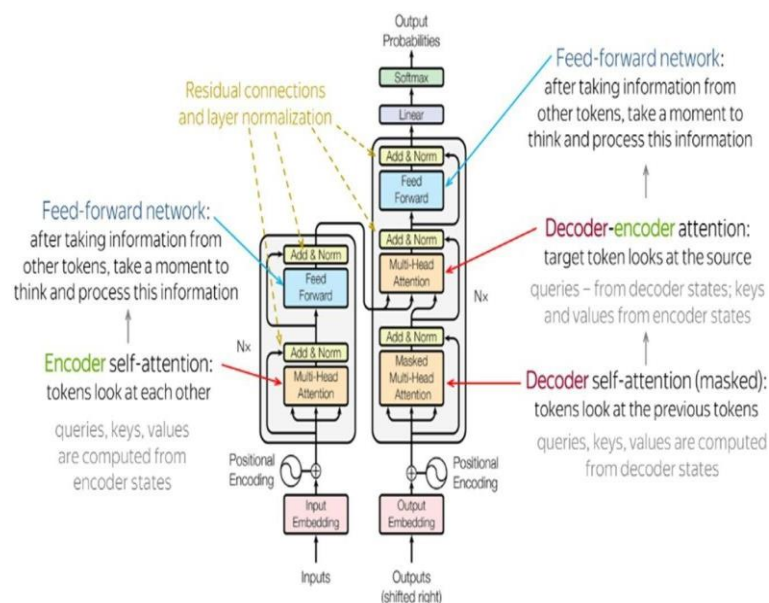


# LARGE LANGUAGE MODEL

07

## LARGE LANGUAGE MODEL (LLM)

- **What is it?**
- Language Models can be defined as a probability distribution over sequences of words. Given a vocabulary of words, a Language Model assigns a probability to each sequence of words, and the purpose is to predict the next word in the sequence. Large Language Models like BERT and GPT are trained on a vast amount of text data to learn the structure, meaning, and usage of language.



source

$$Loss = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_t | x_{<t})$$

VALIDATION

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## VALIDATION BY CONSISTENCY RECORDS

- The consistency test validates the **logical relationships** between three insurance claim variables in the synthetic data:
- ClaimNb** (Number of claims)      **No Claims:** All three = 0 (no claims occurred)
- ClaimOcc** (Claim occurrences)      **Claims Exist:** All three > 0 (claims occurred with positive values)
- ClaimAmount** (Claim amount)

```
# Find inconsistencies
inconsistent_records = new_samples_df[
    ~(((new_samples_df["ClaimNb"] == 0) & (new_samples_df["ClaimOcc"] == 0) & (new_samples_df["ClaimAmount"] == 0)) |
      ((new_samples_df["ClaimNb"] > 0) & (new_samples_df["ClaimOcc"] > 0) & (new_samples_df["ClaimAmount"] > 0)))
]

print(f"Number of inconsistent records: {len(inconsistent_records)}")
print(inconsistent_records.head()) # Show a few inconsistent rows
```

Number of inconsistent records: 1

|       | Exposure    | VehValue | VehAge | VehBody | Gender | DrivAge | ClaimNb | \ |
|-------|-------------|----------|--------|---------|--------|---------|---------|---|
| 49759 | 0.736911    | 5.434924 | 3      | 6       | 1      | 6       | 1.0     |   |
|       |             |          |        |         |        |         |         |   |
|       | ClaimAmount | ClaimOcc |        |         |        |         |         |   |
| 49759 | 0.0         | 1        |        |         |        |         |         |   |

## VALIDATION BY KOLMOGOROV-SMIRNOV TEST

- The Kolmogorov-Smirnov (KS) test compares the distributions of original and synthetic data for each feature. A **high p-value ( $>0.05$ )** indicates that the two distributions are statistically similar, while a **low p-value ( $<0.05$ )** suggests significant differences.

```
]: # Kolmogorov-Smirnov test
for column in X_train.columns:
    original = X_train[column].values
    generated = new_samples_df[column].values
    statistic, p_value = ks_2samp(original, generated)
    print(f"KS Test for {column}: Statistic={statistic}, P-value={p_value}")
```

```
KS Test for Exposure: Statistic=0.13863465866466618, P-value=0.0
KS Test for VehValue: Statistic=0.2830457614403601, P-value=0.0
KS Test for VehAge: Statistic=0.07691297824456114, P-value=1.3001617749392413e-137
KS Test for VehBody: Statistic=0.20864591147786948, P-value=0.0
KS Test for Gender: Statistic=0.001444111027756878, P-value=0.9999999970492804
KS Test for DrivAge: Statistic=0.12738184546136533, P-value=0.0
KS Test for ClaimOcc: Statistic=0.0, P-value=1.0
KS Test for ClaimNb: Statistic=0.0017254313578394243, P-value=0.9999982405003918
KS Test for ClaimAmount: Statistic=0.016185296324081055, P-value=1.6976741222647111e-06
```

# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

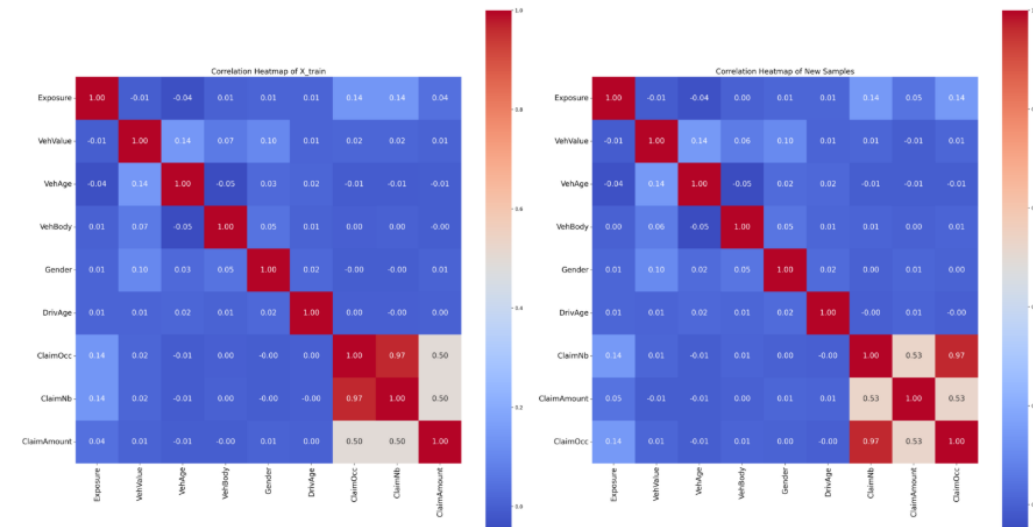
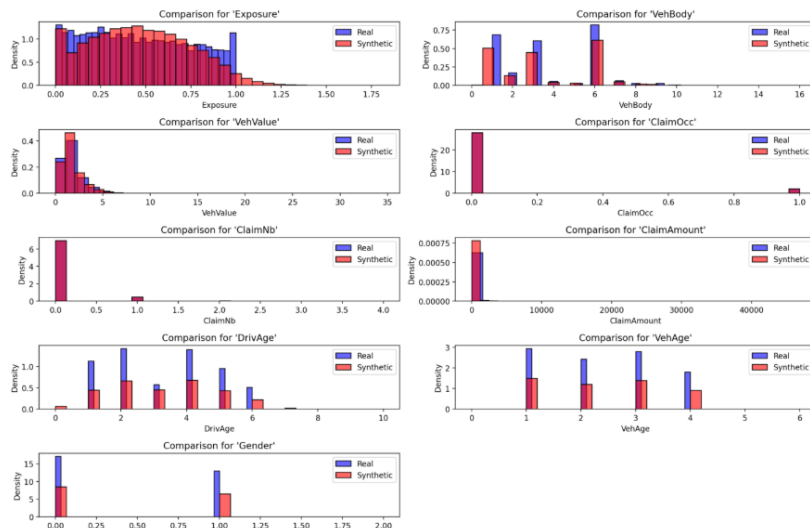
## VALIDATION BY DATA VISUALIZATION

### • Univariate Analysis

- Displays the alignment of the univariate statistical properties for each feature across the real and synthetic data

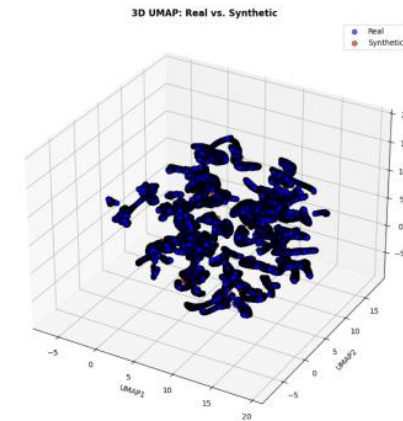
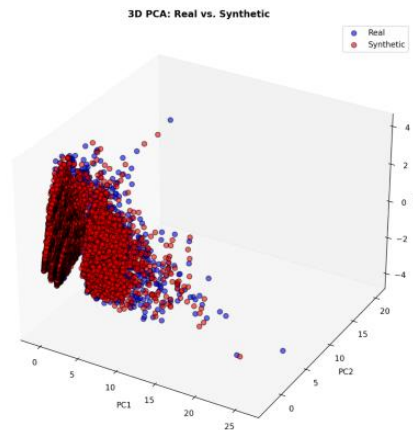
### • Correlation Matrix

- Assesses the structural fidelity of the synthetic data by comparing the correlation matrix to the original data



## VALIDATION BY DATA VISUALIZATION

- **3D PCA**
- Compares the explained variance captured by the principal components of the synthetic data against the real data
- **3D UMAP**
- Evaluates the degree to which the synthetic data preserves the local and global topological structure of the original data



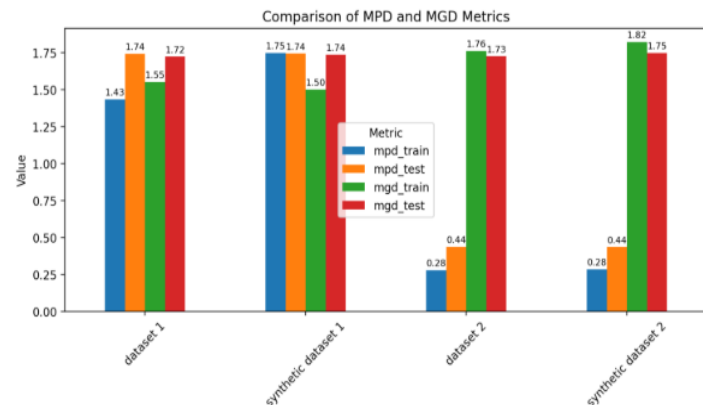


# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## VALIDATION BY PREDICTION AND FEATURE IMPORTANCE

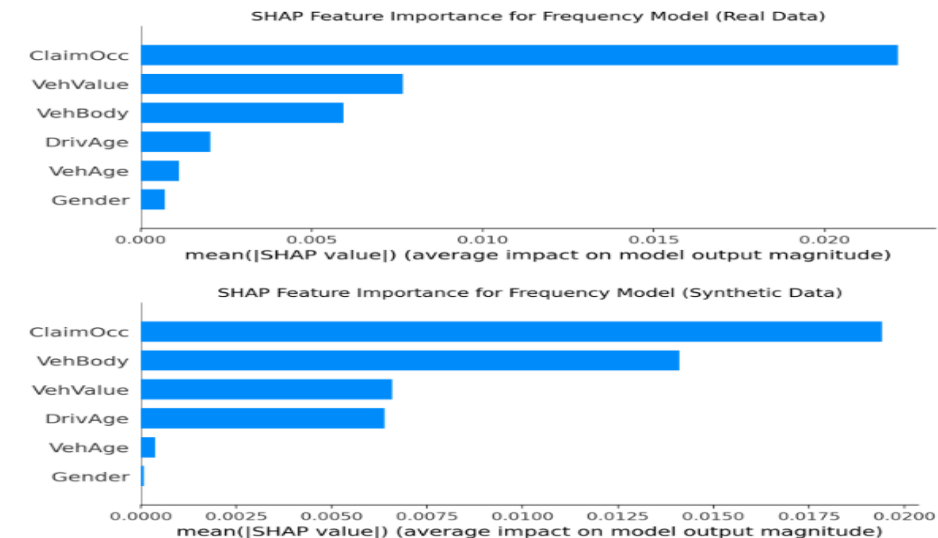
- **Frequency and Severity prediction**
- GLM on both synthetic and real data to evaluate performance using deviance metrics.
- **SHAP Feature Importance**
- Shows whether drivers of frequency and severity predictions remain consistent across datasets.

|                     | mpd_train | mpd_test | mgd_train | mgd_test |
|---------------------|-----------|----------|-----------|----------|
| dataset 1           | 1.4345    | 1.7412   | 1.5509    | 1.7221   |
| synthetic dataset 1 | 1.7457    | 1.7427   | 1.4991    | 1.7361   |
| dataset 2           | 0.2788    | 0.4368   | 1.7615    | 1.7251   |
| synthetic dataset 2 | 0.2848    | 0.4357   | 1.8222    | 1.7478   |



### SHAP Feature Importance for Frequency Model

Shows whether drivers of frequency predictions remain consistent across datasets.



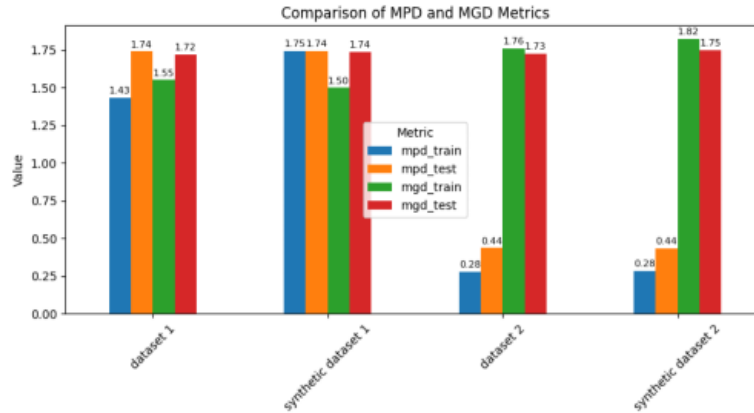
RESULTS

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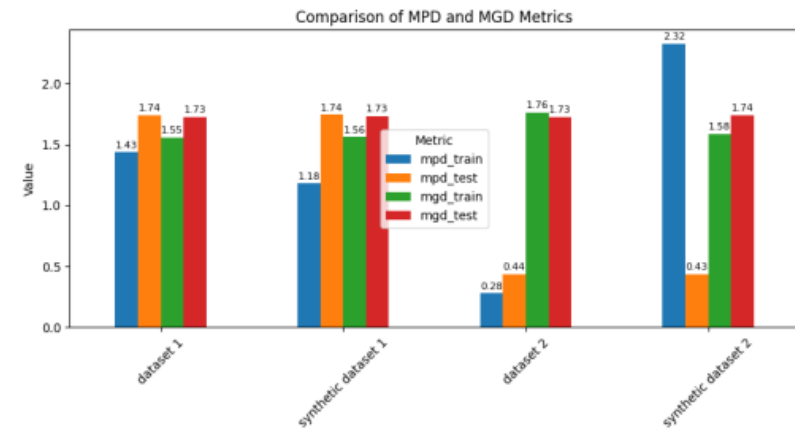
# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## CGMM PREDICTIVE MODELLING RESULTS

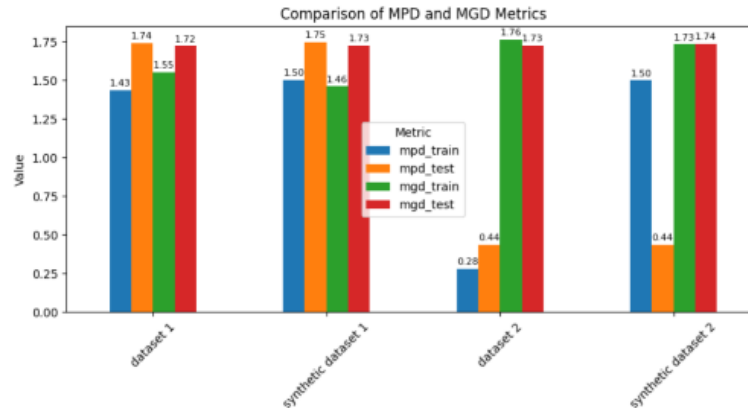
### • Trial 1



### Trial 3



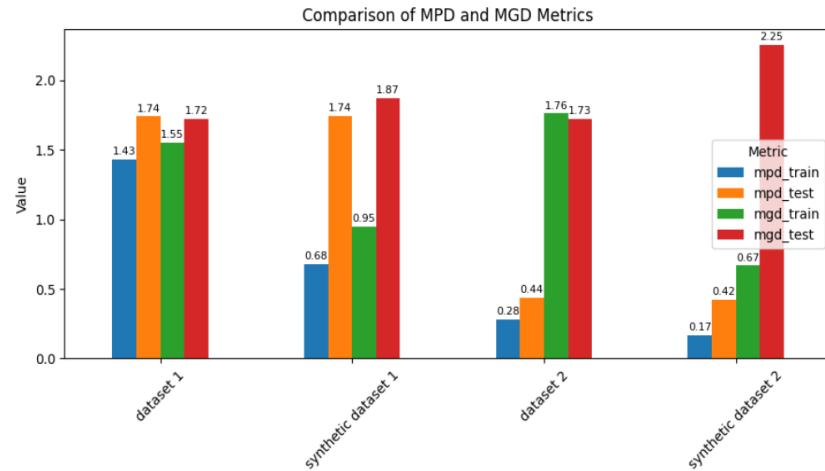
### • Trial 2



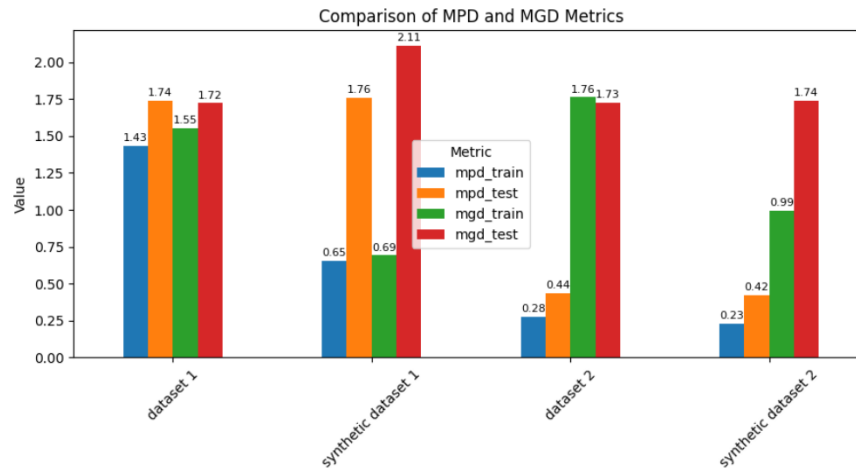
# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## CVAE PREDICTIVE MODELLING RESULTS

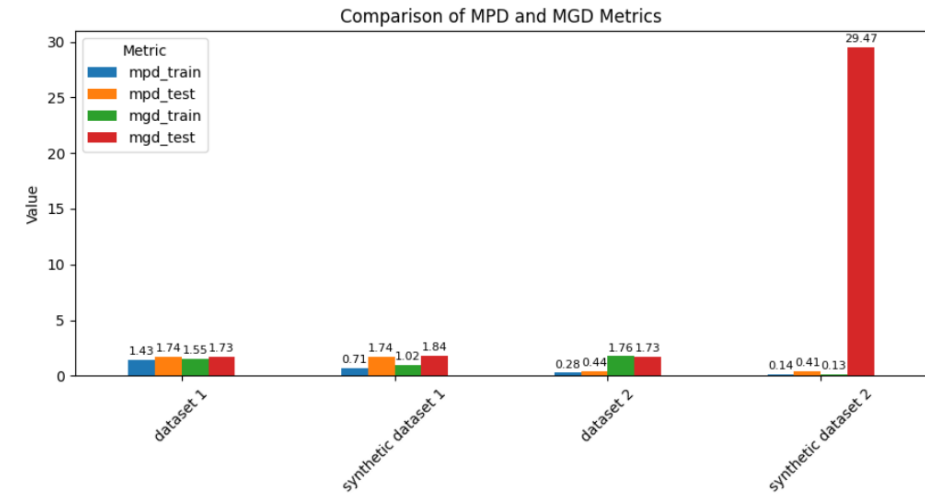
### • Trial 1



### • Trial 2

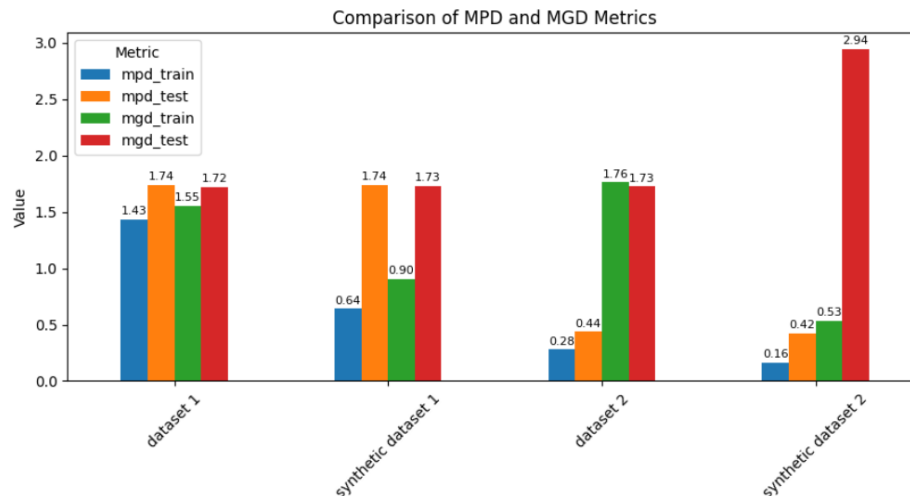


### Trial 3

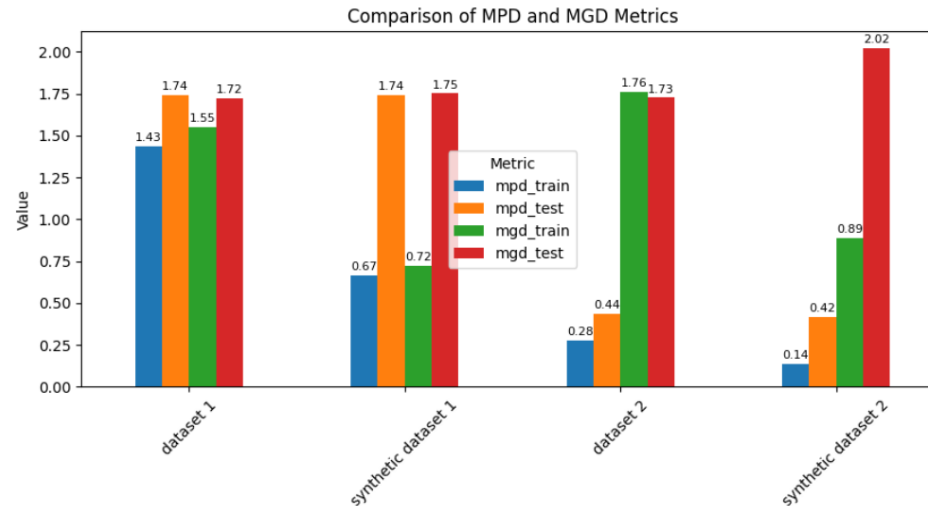


## CTVAE PREDICTIVE MODELLING RESULTS

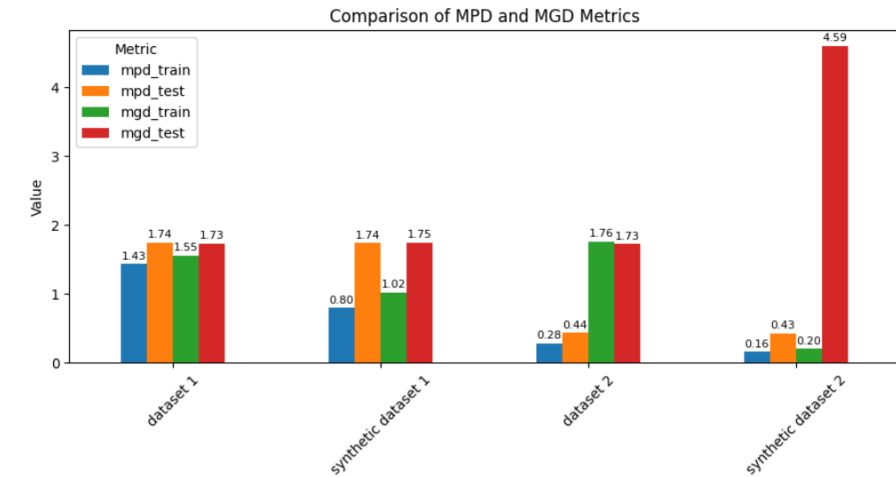
### • Trial 1



### • Trial 2



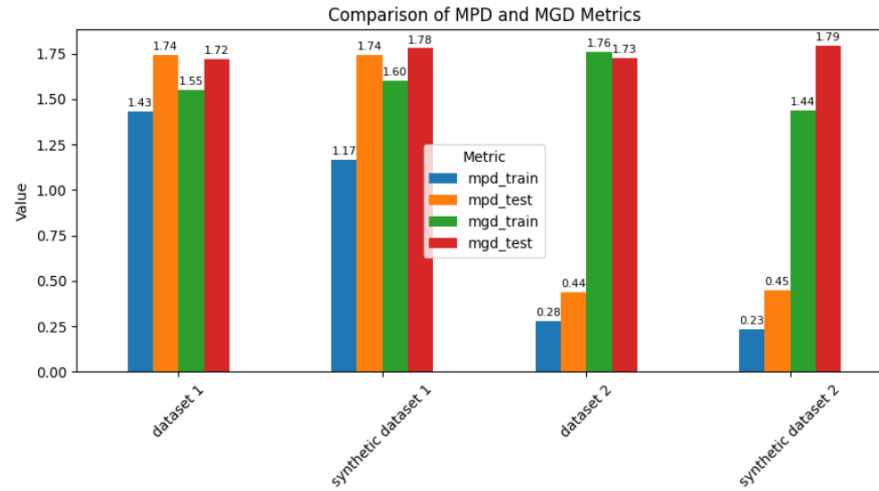
### Trial 3



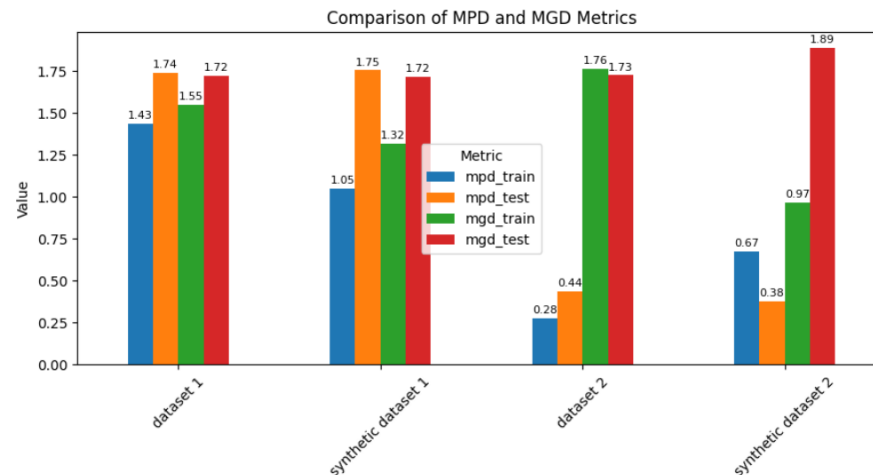
# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## CDM PREDICTIVE MODELLING RESULTS

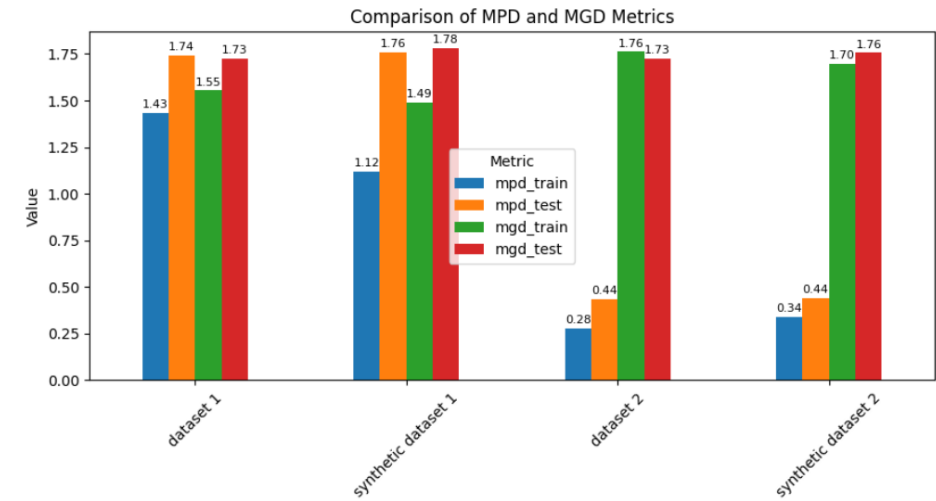
### • Trial 1



### • Trial 2

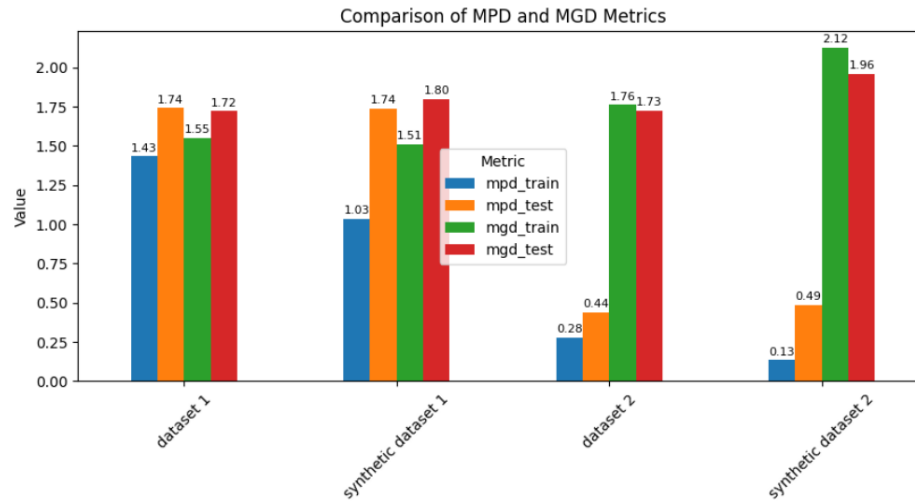


### Trial 3

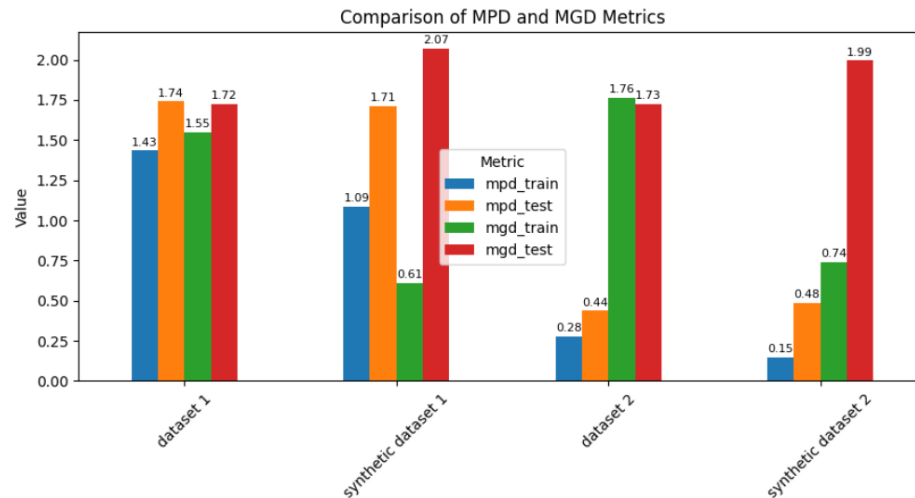


## LLM PREDICTIVE MODELLING RESULTS

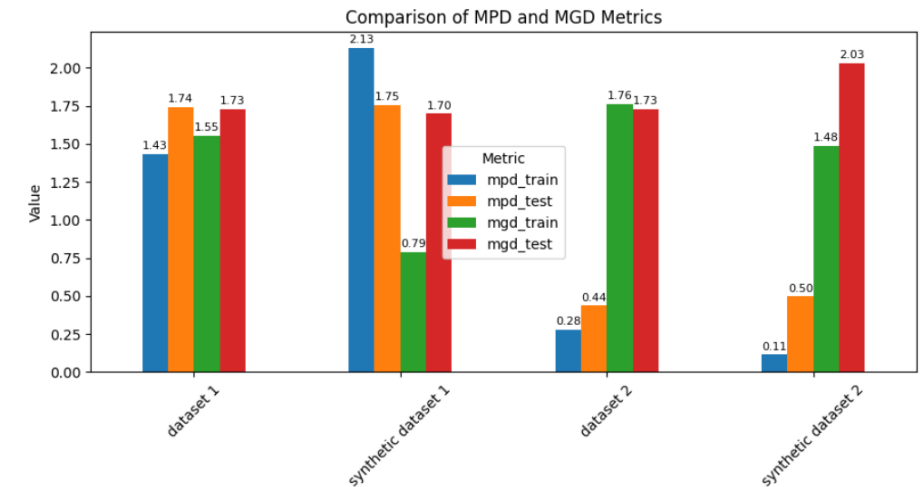
### • Trial 1



### • Trial 2



### Trial 3



## OVERALL RESULTS

- Focusing on predictive modelling performance and the preservation of statistical properties, I recommend the Conditional Gaussian Mixture Model, the Large Language Model, and the Conditional Diffusion Model. The two Conditional Variational Autoencoders show strong training performance, but suffer from severe overfitting. In subsequent trials, model performance degrades, but in some cases it remains acceptable.
- Watch the results from the web demo app.





CONCLUSIONS

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## ***CONCLUSIONS***

-  Conditional Gaussian Mixture Models are competitive as generative models for tabular data. Good compromise between ease of use and effectiveness.
-  Large Language Models are promising for synthetic data generation due to their performance and ease of use. Complex model implementation isn't required, but detailed prompting and deep data analysis are needed to write a good prompt.
-  Conditional Diffusion Model is competitive in performance, but it requires computational and coding effort.

# HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA

## • References

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- -Mario V. Wuthrich, Ronald Richman, Benjamin Avanzi, Mathias Lindholm, Michael Mayer, Jürg Schelldorfer, Salvatore Scognamiglio, 2025, *AI Tools for Actuaries*, SSRN.
- -David Foster, 2023, *Generative Deep Learning, 2nd Edition*, O'Reilly.
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- [Repository](#)
- [WebAPP](#)
- [Article](#)
- [Video](#)

# ***HARNESSING GENERATIVE MODELS FOR SYNTHETIC NON-LIFE INSURANCE DATA***

## Q&A

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# Thank you

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# MySelf:

Actuary during the day &  
AI Scientist in my free time

Reach Me:

MyLinks



## ABOUT ME



Claudio Giorgio  
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Independent

EAA e-Conference on  
Data Science & Data Ethics

20 May 2026



# Thank you very much for your attention

## Contact

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